

CM0081 Formal Languages and Automata

Undecidable Problems

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Semester 2024-1

Introduction

Undecidable Problems

There are undecidable problems in different domains:

- Analysis
- Logic
- Matrices
- Topology
- Physics
- Among other

Normal Forms for the Lambda Calculus

Alonzo Church (1903 – 1995)[†]



[†]Figures sources: [History of computers](#), [Wikipedia](#) and [MacTutor History of Mathematics](#).

Normal Forms for the Lambda Calculus

Some remarks about the λ -calculus

- A formal system invented by Church around 1930s.
- The goal was to use the λ -calculus in the **foundation** of mathematics.
- Intended for studying **functions** and **recursion**.
- Computability model.
- A free-type functional programming language.
- λ -notation (e.g. anonymous functions and currying).

Normal Forms for the Lambda Calculus

Application

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Abstraction

“If M is any formula containing the variable x , then $\lambda x[M]$ is a symbol for the function whose values are those given by the formula.” (Church 1932, p. 352)

Normal Forms for the Lambda Calculus

Currying

“Adopting a device due to Schönfinkel, we treat a function of two variables as a function of one variable whose values are functions of one variable, and a function of three or more variables similarly.” (Church 1932, p. 352)

Such device is called **currying** after Haskell Curry.

(continued on next slide)

Normal Forms for the Lambda Calculus

Currying (continuation)

Let $g : X \times Y \rightarrow Z$ be a function of two variables. We can define two functions f_x and f :

$$f_x : Y \rightarrow Z$$

$$f_x = \lambda y. g(x, y),$$

$$f : X \rightarrow (Y \rightarrow Z)$$

$$f = \lambda x. f_x.$$

Then $(f\ x)\ y = f_x\ y = g(x, y)$. That is, the function of two variables

$$g : X \times Y \rightarrow Z$$

is represented as the higher-order function

$$f : X \rightarrow (Y \rightarrow Z).$$

Normal Forms for the Lambda Calculus

Definition

Let V be a denumerable set of variables. The set of **λ -terms**, denoted by Λ , is inductively defined by

$$x \in V \Rightarrow x \in \Lambda \quad \text{(variable)}$$

$$M, N \in \Lambda \Rightarrow (MN) \in \Lambda \quad \text{(application)}$$

$$M \in \Lambda, x \in V \Rightarrow (\lambda x.M) \in \Lambda \quad \text{(\(\lambda\)-abstraction)}$$

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Observation

Usually, the set of λ -terms is defined by an abstract grammar like

$$t ::= x \mid tt \mid \lambda x.t$$

Normal Forms for the Lambda Calculus

Conventions

- λ -term **variables** will be denoted by $x, y, z, \dots$.
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Example

Whiteboard.

Normal Forms for the Lambda Calculus

Conventions and syntactic sugar

- Outermost parentheses are not written.
- Application has higher precedence, i.e.,

$$\lambda x.MN := (\lambda x.(MN)).$$

- Application associates to the left, i.e.,

$$MN_1N_2 \dots N_n := (\dots ((MN_1)N_2) \dots N_n).$$

- Abstraction associates to the right, i.e.,

$$\begin{aligned}\lambda x_1x_2 \dots x_n.M &:= \lambda x_1.\lambda x_2. \dots \lambda x_n.M \\ &:= (\lambda x_1.(\lambda x_2.(\dots (\lambda x_n.M) \dots))).$$

Normal Forms for the Lambda Calculus

Definition

A variable x occurs **free** in M if x is not in the scope of λx . Otherwise, x occurs **bound**.

Definition

The **set of free variables in M** , denoted by $FV(M)$, is inductively defined by

$$\begin{aligned}FV(x) &:= \{x\}, \\FV(MN) &:= FV(M) \cup FV(N), \\FV(\lambda x.M) &:= FV(M) - \{x\}.\end{aligned}$$

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Notation

The symbol “ $\equiv$ ” denotes the syntactic identity.

Normal Forms for the Lambda Calculus

Definition

The result of **substituting N for every free occurrence of x in M** , and changing bound variables to avoid clashes, denoted by $M[x/N]$, is defined by (Hindley and Seldin 2008, Definition 1.12)

$$x[x/N] := N,$$

$$y[x/N] := y, \text{ if } y \neq x,$$

$$(PQ)[x/N] := (P[x/N] Q[x/N]),$$

$$(\lambda x.P)[x/N] := \lambda x.P,$$

$$(\lambda y.P)[x/N] := \lambda y.P, \text{ if } y \neq x \text{ and } x \notin \text{FV}(P),$$

$$(\lambda y.P)[x/N] := \lambda y.P[x/N], \text{ if } y \neq x, x \in \text{FV}(P) \text{ and } y \notin \text{FV}(N),$$

$$(\lambda y.P)[x/N] := \lambda z.P[x/N][y/z], \text{ if } y \neq x, x \in \text{FV}(P) \text{ and } y \in \text{FV}(N),$$

where in the last equation, the variable z is chosen such that $z \notin \text{FV}(NP)$.

Normal Forms for the Lambda Calculus

Definition

The functional behaviour of the λ -calculus is formalised through of their reduction/conversion rules. The β -**reduction rule** is defined by

$$(\lambda x.M)N \rightarrow_{\beta} M[x/N].$$

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- $(\lambda y.yy)x \rightarrow_{\beta} xx$
- $(\lambda x.(\lambda y.yx)z)v \rightarrow_{\beta} (\lambda y.yv)z \rightarrow_{\beta} zv$
- Let Ω be $(\lambda x.xx)(\lambda x.xx)$, then $\Omega \rightarrow_{\beta} \Omega \rightarrow_{\beta} \dots$

Normal Forms for the Lambda Calculus

Definition

A β -**redex** is a λ -term of the form $(\lambda x.M)N$.

Definition

A λ -term which contains no β -redex is in β -**normal form** (β -nf).

Definition

A λ -term N is a β -**nf of** M (or M **has the** β -**nf** M) iff N is a β -nf and $M =_{\beta} N$, where $=_{\beta}$ is the equivalence relation generated by the reflexive and transitive closure of $\rightarrow_{\beta}$.

Example

Whiteboard.

Normal Forms for the Lambda Calculus

Theorem

The set

$$\text{NF} := \{ M \in \Lambda \mid M \text{ has normal form} \}$$

is not recursive (i.e. it is undecidable) (Church 1935, 1936).

Observation

This was the **first** undecidable set ever.

Normal Forms for the Lambda Calculus

Observation

For proving that the set **NF** is undecidable we need an encoding and a version of Rice's theorem for λ -calculus.

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Gödel numbering

The Gödel numbering for the λ -terms is defined by

$$\begin{aligned}\# : \Lambda &\rightarrow \mathbb{N} \\ \#(x_i) &= 2^i, \\ \#(\lambda x_i.M) &= 3^i 5^{\#(M)}, \\ \#(MN) &= 7^{\#(M)} 11^{\#(N)}.\end{aligned}$$

Normal Forms for the Lambda Calculus

Theorem (Rice's theorem for the λ -calculus)

Let $A \subset \Lambda$ such as A is non-trivial (i.e. $A \neq \emptyset$ and $A \neq \Lambda$). Suppose that A is closed under $=_\beta$ (i.e. $M \in A$ and $M =_\beta N$ then $N \in A$). Then the set A is undecidable, that is,

$$\{ \#(M) \mid M \in A \} \text{ is undecidable.}$$

See (Barendregt 1992).

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See (Barendregt 1992).

Proof (undecidability of NF)

Since the set NF is not trivial and it is closed under $=_\beta$, the set is undecidable. ■

The *Entscheidungsproblem*

The problem

The *Entscheidungsproblem* (decision problem) can be stated in three equivalent ways (Davis 2013, p. 49):

- (i) “Find an algorithm to determine whether a given sentence of first order logic is valid, that is, true regardless of what specific objects and relationships are being reasoned about.”
- (ii) “Find an algorithm to determine whether a given sentence of first order logic is satisfiable, that is, true for some specific objects and relationships.”
- (iii) “Find an algorithm to determine given some sentences of first order logic regarded as premises and another sentence, being a desired conclusion, whether that conclusion is provable from the premises using the rules of proof for first order logic.”

The *Entscheidungsproblem*

Historical remark

The *Entscheidungsproblem* was posed by Hilbert and Ackermann in 1928 (Hilbert and Ackermann 1950).

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Negative answer

Church (1935, 1936) and Turing (1936–1937) gave a negative answer to the *Entscheidungsproblem* from the undecidability of the normal forms for the λ -calculus and the halting problem for Turing machines, respectively.

Post's Correspondence Problem (PCP)

An instance of the PCP

An instance of PCP consist of two lists of equal length

$$A = w_1, \dots, w_k \quad \text{and} \quad B = x_1, \dots, x_k$$

of strings over an alphabet Σ .

(continued on next slide)

Post's Correspondence Problem (PCP)

An instance of the PCP (continuation)

We say that the previous instance of PCP **has a solution**, if there is a sequence of one or more integers

$$i_1, \dots, i_m, \text{ with } m \geq 1$$

that, when interpreted as indexes for strings in the A and B lists, yield the same string, i.e.

$$w_{i_1} \cdots w_{i_m} = x_{i_1} \cdots x_{i_m}.$$

The sequence

$$i_1, \dots, i_m$$

is called a **solution** of the instance of PCP.

Post's Correspondence Problem (PCP)

The problem

Given an instance of PCP, tell whether this instance has a solution.

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Example 9.13

An instance of the PCP:

	List A	List B
i	w_i	x_i
1	1	111
2	10111	10
3	10	0

Solution: $2, 1, 1, 3, m = 4$.

Post's Correspondence Problem (PCP)

Undecidability proof

The PCP problem is undecidable (Post 1946). Hopcroft, Motwani and Ullman (2007) shows the undecidability via a reduction of L_u to PCP.

The Mortal Matrix Problem (MMP)

The problem

Let S be a finite set of $n \times n$ matrices with integer entries. To determine whether the zero matrix belongs to the semigroup generated by S , i.e. to determine whether the matrices in S can be multiplied in some order, possibly with repetitions, to yield the zero matrix.

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Some undecidable instances

The MMP is undecidable for a set of seven 3×3 matrices, or a set of two 21×21 matrices (Halava, Harju and Hirvensalo 2007).

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Undecidability proof

Reduction of PCP to MMP.

Hilbert's Tenth Problem

Definition

A **Diophantine equation** is an equation of the form

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The problem (in present terminology)

*“Given a Diophantine equation with any number of unknowns: **To devise a process** according to which it can be determined by a **finite** number of operations whether the equation has non-negative integer solutions.” (Sicard, Ospina and Vélez 2006, p. 12542)*

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Undecidability proof

A set is recursively enumerable if and only if it is Diophantine (Matiyasevich 1993).

Undecidable Problems in Physics

Some undecidable problems

“Physics is also full of non-computable problems. The undecidability of the presence of chaos in classical Hamiltonian systems has been established ³³. The problem whether a boolean combination of subspaces (including negations) is reachable by a quantum automation was proved to be undecidable ³⁴. The question whether a quantum system is gapless also cannot be decided by an algorithm ^{35–37}. Whether a many-body model is frustration-free is undecidable as well ³⁸. Smith (Sec. 6 of ³⁹) identified a striking physical consequence of the Hilbert’s tenth problem that ground state energies and half-life times of excited states are, strictly speaking, non-computable for many-body systems. A variety of seemingly simple problems in quantum information theory has been shown not to be decidable ⁴⁰. The question whether a sequence of outcomes of some sequential measurement cannot be observed is undecidable in quantum mechanics, whereas it is decidable in classical physics ⁴¹. In this case, the algorithmic undecidability turned out to be the signature of quantumness.” (Bondar and Pechen 2020, p. 2)

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